

CS101: Problem Solving with Algorithms & CS101L: Programming Lab

Spring 2026. Instructor: Brent Magnusson, magnussonb@arcadia.edu

Time: Tuesday/Thursday 4-6:05pm

Location: Heights Philadelphia, 123 S Broad St #850, Philadelphia, PA 19109

Tentative Schedule

1/13 - Introductions, set up accounts, Lab1
1/15 - Lab1-Command Line
1/20 -Touch Typing Initial Test
1/22 - **Quiz**; Lab2-PrintStatements
1/27 - Variables
1/29 - **Quiz**
2/3 - Input/Output
2/5 - **Quiz**
2/10 - Conditionals
2/12 - **Quiz**
2/17 - NO CLASS
2/19 -
2/24 - For Loops
2/26 - **Quiz**
3/3 - While Loops
3/5 - **Quiz**
3/10 - Methods
3/12 - **Quiz**
3/17 - **Touch Typing Midterm Test**
3/19 - **Quiz**
3/24 -
3/26 - NO CLASS
3/31 - NO CLASS
4/2 - NO CLASS
4/7 -
4/9 - **Quiz**; Arrays
4/14 -
4/16 - **Quiz**
4/21 -
4/23 - **Quiz**
4/28 -
4/30 - **Quiz**
5/5 -
5/7 - **Quiz**
5/12 -
5/14 - **Final Exam**
5/19 - NO CLASS
5/21 - **Final Project Demos; Touch Typing Final Test**

Resources

Office Hours: Group Zoom Office Hours 4-5:30pm on Fridays. Individual Zoom by appointment (email me).

Canvas: Home base for class announcements, grades, and submitting labs/assignments

GitHub CS101 Page: Location for topic resources, labs, and assignments

Learning Objectives

Technical: By the end of the course, you will be able to leverage the following concepts in Java programs:

1. Variables
2. Input/output from user and files
3. Conditional statements (if, elseif)
4. Loops (for, while, do-while)
5. Methods
6. Arrays

General: Throughout the course, you will practice and develop your ability in the following domains:

1. Structured Thinking (breaking down complex problems into smaller steps)
2. Agency and self-directed learning
3. Patience and persistence during problem-solving

Grade Assessment

10% - Attendance

25% - Labs

18% - Homework Assignments

20% - Quizzes (2% each)

7% - Final Exam

20% - Final Project

4% (extra credit) - Touch Typing

F	D-	D	D+	C-	C	C+	B-	B	B+	A-	A
0-59	60	63	67	70	73	77	80	83	87	90	93

Grades are rounded to 1 decimal place and are left side inclusive, right side exclusive. ie: a final grade of 86.94 will be rounded to 86.9, which is a B. A final grade of 86.95 will be rounded to 87.0, which is a B+.

Attendance

You must arrive by 4:05pm and stay through the entire class to be counted as present. You are allowed 3 missed class days (excused or unexcused). Additional missed days will reduce your overall grade by 1% each, up to a maximum penalty of 10%. ie: if you miss 3 days, your maximum final grade is 100%. If you miss 5 days, it is 98%. If you miss 15 days, your max possible grade is 90%. This penalty may be reduced if communication about future absences is done in a timely fashion.

Homework Assignments

Assignments must be turned in by midnight on the day they are due. Every additional 24 hours that an assignment is late (excluding Weekends and Holidays) will reduce the maximum possible points by 5%. For the entire semester, you are allotted 3 late days with no penalty.

Quizzes

There will be a quiz most Thursdays at the beginning of class. If you miss class that day, you will receive a score of zero on that quiz. **YOU CANNOT MAKEUP A QUIZ!** Your worst 3 quiz scores will be dropped at the end of the semester. There will be 13 quizzes, meaning that your best 10 scores will count, and that each quiz is worth 2% of your total grade.

Labs

Exercises done either individually or in small groups during class time or at home. Can be submitted multiple times. You will receive a zero (and get feedback from me) until the assignment is deemed 100% correct. Typically due 3 weeks after being assigned. Labs can be submitted late, up to the day of the final. Late labs will only receive 50% of the total points (and are still required to be completely correct to get any credit).

Final Exam

The final will cover material from the entire semester. **Attendance at the final is mandatory to pass the class.** If you have an emergency and won't make the final, you must email me **before** the scheduled time of the exam.

Final Project

Details forthcoming.

Touch Typing (Extra Credit)

We will do initial typing speed tests in the first week. There will be a typing test to measure progress around midterm time, worth up to 2 extra credit percentage points towards your overall final grade. Another test will be done at the end of the semester, again worth up to 2 extra credit points. The metric used to measure progress is TBD.

Academic Honesty

For programming assignments (labs, homeworks, final project), cheating is defined as copy-pasting any amount of code without understanding how it works and what it does. ie:

- Copying an implementation from a classmate
- Copying code from Stack Overflow, Reddit, etc, without proper understanding
- Using AI to generate code for you

For any line of code you submit in this class, I may ask you to explain what that line is doing in words. If you are unable to explain it, I will assume that the line was blindly copy-pasted.

Based on [Arcadia University Regulations and Policies](#), for quizzes and exams, cheating is defined as "Attempting to look at another student's exam. Attempting to communicate concerning the content of the exam with another student. Attempting to use any materials (such as notebooks, notes, textbooks) not specifically authorized by the faculty member."

If I determine that you cheated on a quiz, homework, or exam, you will receive a zero for that assignment. You will also receive a zero if you knowingly allow a fellow student to copy your work. If you disagree with the penalty, you are entitled to submit a formal request to the university for review. See the section titled "Procedures for Reporting Violations and Determining Responsibility" in the [Arcadia University Regulations and Policies](#).